

Evaluation Report — 2026-06-25

A side-by-side comparison of four AI coding tools run against an identical set of prompts. Each tool was measured on **cost**, **speed**, and **correctness** across three scenarios: one greenfield build and two brownfield analysis tasks.

Summary

- **Run date:** 2026-06-25
- **Aggregation:** mean over 1 run per tool/scenario
- **Mode:** Automated
- **Tools under test:** Claude Code, Codex CLI, Kiro CLI, Kiro IDE

Headline findings

- **Lowest cost:** Kiro CLI at **\$0.1348** total across the three scenarios — roughly 26x cheaper than Codex CLI and 14x cheaper than Claude Code.
- **Fastest:** Kiro CLI completed all three scenarios in **220.5s** total (73.5s average), the quickest of the group.
- **Most expensive:** Codex CLI at **\$3.5473** total, driven by large input token counts billed at the GPT-5.5 rate.
- **Best cost/speed balance:** Kiro CLI — the cheapest and the fastest tool in this run. Kiro IDE was the second-cheapest (\$0.1520 total) at credit-based pricing close to Kiro CLI.

Configuration

Tool	Model	Effort	Mode
Claude Code	us.anthropic.claude-opus-4-8	xHigh	Automated
Codex CLI	openai.gpt-5.5	xHigh	Automated
Kiro CLI	claude-opus-4.8	xHigh	Automated
Kiro IDE	claude-opus-4.8	xHigh	Manual

Methodology

- **Harness:** runs were executed and aggregated by `tools/run_benchmark.py` (run-average mode) and recorded in `run-complete.json`.
- **Procedure:** each tool received the identical prompt for each scenario, run in Automated mode with no manual intervention. Output was written to the tool's designated folder and metrics captured from each tool's own usage reporting.
- **Runs and aggregation:** 1 run per tool/scenario; reported figures are the mean (a single run here, so mean equals the observed value).
- **Metrics captured:**
 - *Cost (USD)* — derived per the pricing and methodology below.
 - *Time (s)* — wall-clock elapsed time from prompt to finished result.

- *Lines* — lines of output produced (where reported by the tool).
- *Tokens* — input/output token counts (where reported by the tool).
- **Held constant:** prompts, scenario order, model effort level (xHigh across all tools), and Automated mode across all tools.
- **Not scored:** correctness/quality of output is out of scope for this run and assessed separately.

Pricing

OpenAI GPT 5.5 (on Amazon Bedrock) — used for Codex CLI

Token type	Price per 1M tokens
Input	\$5.50
Cached input	\$0.55
Output	\$33.00

Claude Opus 4.8 (on Amazon Bedrock) — used for Claude Code

Token type	Price per 1M tokens
Input	\$5.00
Output	\$25.00

Kiro credits — used for Kiro CLI and Kiro IDE

Unit	Price
Per credit	\$0.02

Cost methodology

- **Kiro CLI / Kiro IDE:** derived from credits consumed at **\$0.02/credit**.
- **Claude Code:** reported run cost in USD from the `/usage` command (Connected to Amazon Bedrock, Claude Opus 4.8 at \$5.00/1M input and \$25.00/1M output; reported totals also include cache read/write).
- **Codex CLI:** computed from session token totals using the GPT-5.5 on Amazon Bedrock pricing above (input × \$5.50/1M + output × \$33.00/1M).

Evaluated Scenarios

- **GF-1** — Greenfield build: produce a fresh, spec-bound CLI to-do app from scratch.
- **CA-1** — Brownfield analysis: read an existing TypeScript API and explain architecture, pricing, and the order state machine.
- **CA-2** — Brownfield investigation: pinpoint which check rejects a specific request and what response it returns.

Results

Full results matrix

Scenario	Tool	Model	Effort	Cost (USD)	Time (s)	Lines	Input Tokens	Output Tokens
GF-1	Claude Code	Claude Opus 4.8	xHigh	\$0.3352	66.3	209	2,311	6,257
GF-1	Codex CLI	GPT-5.5	xHigh	\$0.9337	112.6	154	128,130	6,940
GF-1	Kiro CLI	Claude Opus 4.8	xHigh	\$0.0308	46.6	203	NA	NA
GF-1	Kiro IDE	Claude Opus 4.8	xHigh	\$0.0540	138.0	NA	NA	NA
CA-1	Claude Code	Claude Opus 4.8	xHigh	\$0.9711	256.5	330	2,446	22,590
CA-1	Codex CLI	GPT-5.5	xHigh	\$1.5280	131.5	211	229,972	7,974
CA-1	Kiro CLI	Claude Opus 4.8	xHigh	\$0.0646	121.4	269	NA	NA
CA-1	Kiro IDE	Claude Opus 4.8	xHigh	\$0.0620	194.0	NA	NA	NA
CA-2	Claude Code	Claude Opus 4.8	xHigh	\$0.6179	135.7	112	2,450	11,602
CA-2	Codex CLI	GPT-5.5	xHigh	\$1.0856	108.2	132	159,971	6,234
CA-2	Kiro CLI	Claude Opus 4.8	xHigh	\$0.0394	52.5	87	NA	NA
CA-2	Kiro IDE	Claude Opus 4.8	xHigh	\$0.0360	53.0	NA	NA	NA

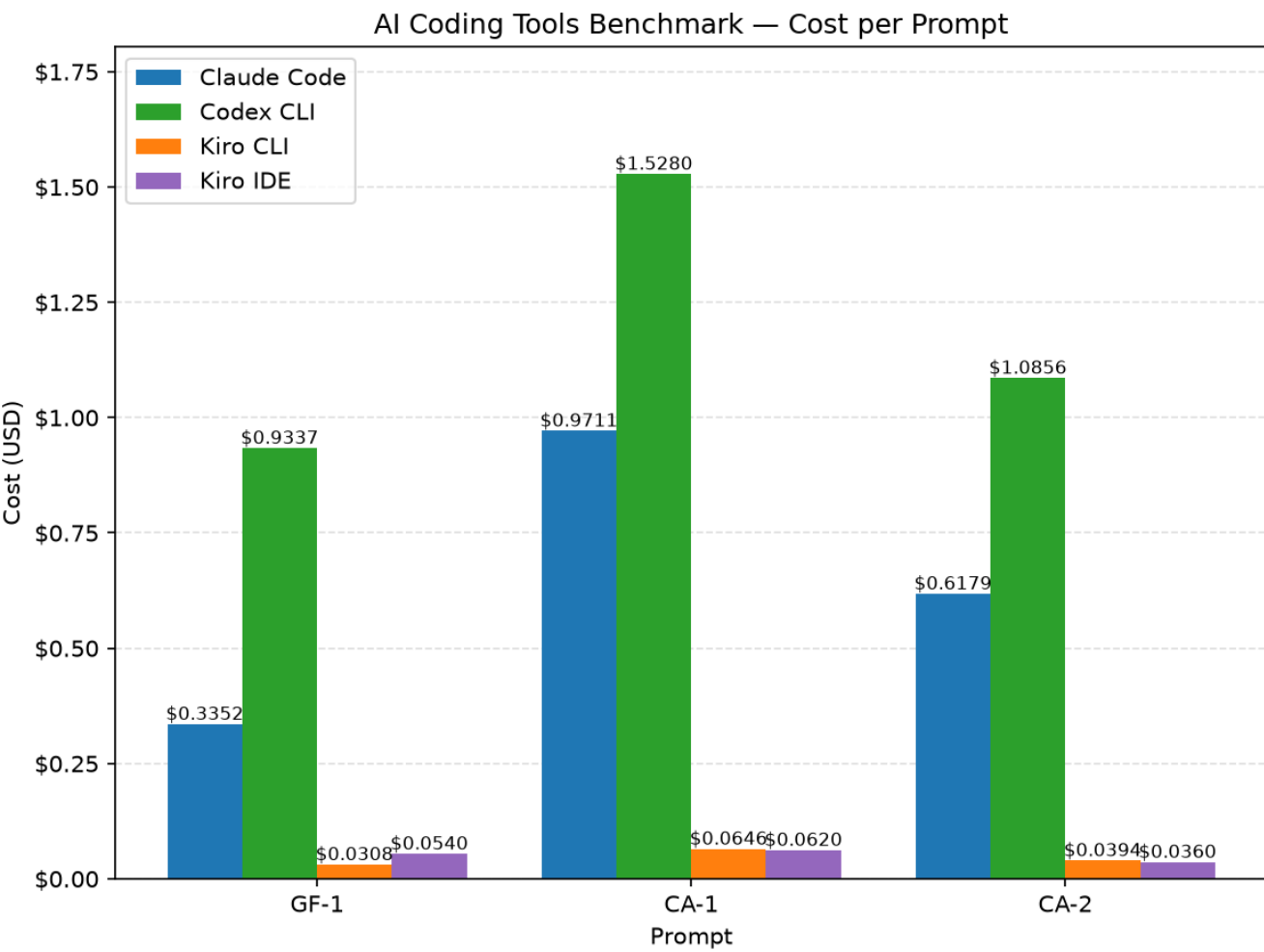
Token counts are not reported for Kiro CLI/IDE (credit-based billing). Kiro IDE does not report a line count in this run.

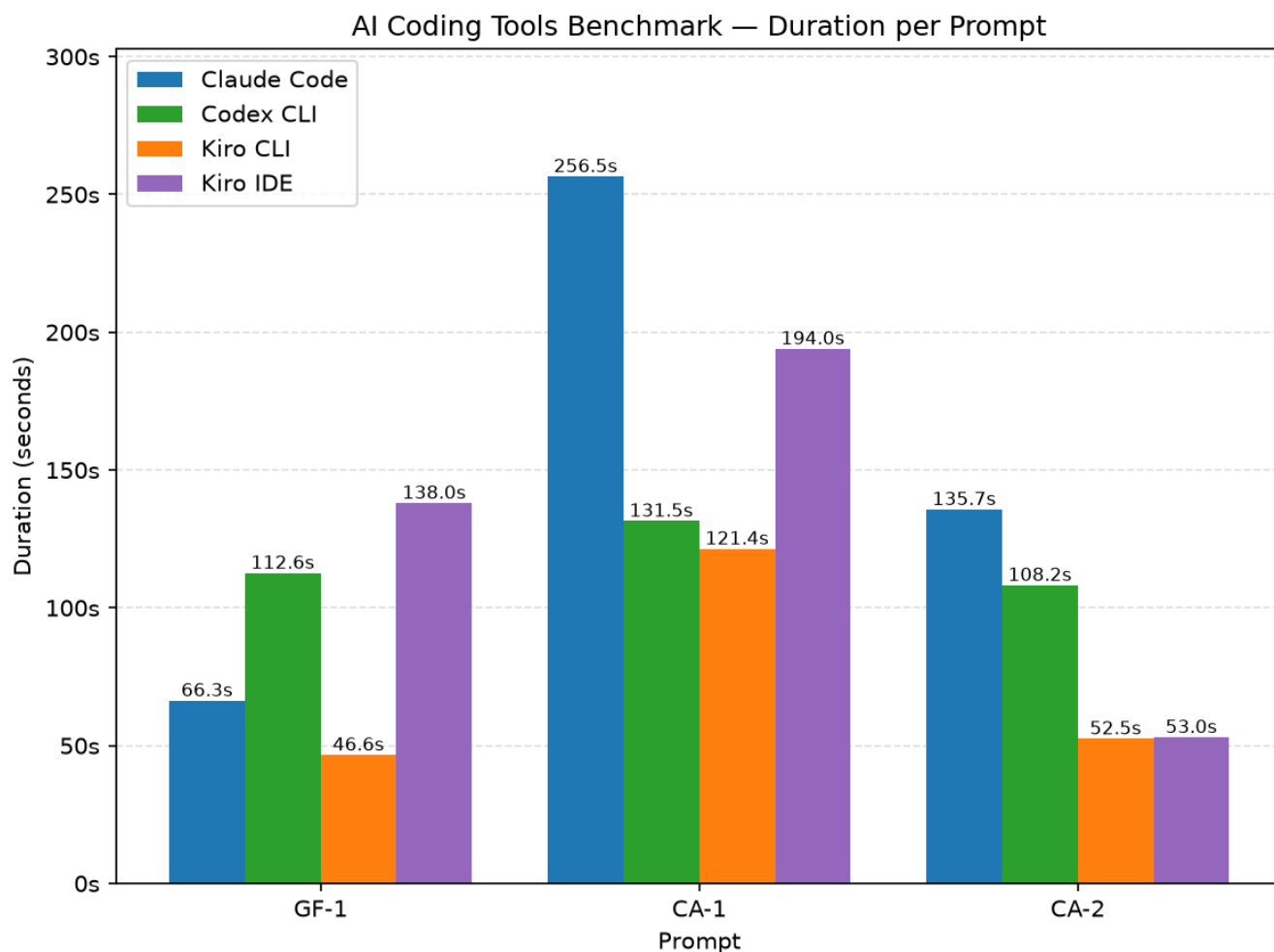
Aggregate by tool

Totals and means across all three scenarios:

Tool	Total Cost (USD)	Avg Cost (USD)	Total Time (s)	Avg Time (s)	Total Lines
Kiro CLI	\$0.1348	\$0.0449	220.5	73.5	559
Kiro IDE	\$0.1520	\$0.0507	385.0	128.3	NA
Claude Code	\$1.9242	\$0.6414	458.5	152.8	651
Codex CLI	\$3.5473	\$1.1824	352.3	117.4	497

Visualizations





Per-Scenario Breakdown

GF-1 — Greenfield build

Output volume was comparable between Claude Code (209 lines) and Kiro CLI (203 lines), with Codex CLI lower at 154. On cost, Kiro CLI (\$0.0308) and Kiro IDE (\$0.0540) were an order of magnitude cheaper than Claude Code (\$0.3352) and Codex CLI (\$0.9337). Kiro CLI was also the fastest (46.6s). Codex CLI's large input token footprint (128,130 tokens) made it the most expensive on this scenario.

CA-1 — Brownfield analysis

This was the most demanding scenario and the widest cost spread. Codex CLI cost \$1.5280 — the single most expensive result in the run — driven by 229,972 input tokens. Claude Code followed at \$0.9711 with the highest output token count (22,590). Kiro CLI delivered the analysis for \$0.0646 (roughly 24x cheaper than Codex CLI) and was the fastest at 121.4s, with Kiro IDE close behind on cost (\$0.0620) in 194.0s.

CA-2 — Brownfield investigation

The lightest scenario by output. Kiro CLI was both the cheapest (\$0.0394) and the fastest (52.5s), with Kiro IDE essentially matched on both (\$0.0360, 53.0s). Codex CLI was again the most expensive among the token-billed tools (\$1.0856), with Claude Code in the middle at \$0.6179.

Observations

- **Credit-based tools dominate on cost.** Kiro CLI and Kiro IDE were consistently 10–30x cheaper per scenario than the token-billed tools, because Codex CLI in particular sends very large input contexts (over 100K tokens per scenario) billed at \$5.50/1M.
- **Kiro CLI led on both cost and speed.** It was the cheapest tool overall (\$0.1348 total) and the fastest (73.5s average), reversing the cost/speed tradeoff seen in earlier runs.
- **Claude Code's cost scales with output volume** — CA-1's 22,590 output tokens pushed it to \$0.9711, its most expensive scenario.
- **Codex CLI was the most expensive in every scenario**, driven by its large input token counts rather than output.

Caveats

- Results reflect a **single run** per tool/scenario; figures are means but not yet variance-tested. Treat individual numbers as indicative rather than definitive.
- Kiro token counts (CLI and IDE) and Kiro IDE line counts are unavailable due to how each tool reports usage, limiting like-for-like comparison on those dimensions.
- This report measures cost, speed, and output volume only. **Correctness of the output is not scored here** and should be assessed separately before drawing quality conclusions.

References

- Source repository: <https://github.com/aws-samples/sample-ai-coding-tools-benchmark>